

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 05)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

- A Its total annual discharge ☐
- B The albedo of its surface ☐
- C Its salinity ☐
- D The age of its bedrock ☐

0 1 . 2

Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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0 1 . 3

Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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5

0 2 . 1

Give **three** advantages of incinerating domestic waste.

[3 marks]

1

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2

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3

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0 2 . 2

Explain how each of the following features of a modern incinerator reduces pollution.

[4 marks]

High combustion temperature with rapid cooling of gases

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Water spray scrubber

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Removal of wet / compostable waste

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Removal of recyclable waste

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0 2 . 3

Explain why the ash from waste incinerators needs careful disposal.

[2 marks]

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0 2 . 4

State **one** reason why incineration is expensive.

[1 mark]

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10

0 3 . 1

Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence

☐

B Low liposolubility and low persistence

☐

C High liposolubility and low persistence

☐

D Low liposolubility and high persistence

☐

0 3 . 2

Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

03 . 3

Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

5

04 . 1

Explain how each of the following human activities changes the amount of carbon in the atmosphere.

[6 marks]

Ploughing farmland

Draining marshland / peat bogs

Toxic pollution of the sea

04 . 2

Give **three** anaerobic environments created by human activities that release methane.

[3 marks]

1

2

3

0 4 . 3 Explain how rising atmospheric CO₂ causes ocean acidification.

[1 mark]

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10

0 5 In the 1950s a chemical factory at Minamata, Japan, discharged inorganic mercury compounds into a shallow bay. Thousands of local people who ate fish and shellfish from the bay suffered severe neurological damage and many babies were born with birth abnormalities.

0 5 . 1 Explain how the low-toxicity inorganic mercury discharged into the bay came to cause such severe poisoning.

[4 marks]

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0 5 . 2 Explain why methyl mercury caused both neurological damage in adults and birth abnormalities.

[3 marks]

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05 . 3

Compare the toxicity of the three chemical forms of mercury.

[3 marks]**10**

06

Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	1	2	4	6	8	10	12	14
Running mean catch per night	31	18	24	21	22	22	21	22

06 . 1

Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

06 . 2

Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

06 . 3

Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

06 . 4

Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

10

07 . 1

Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

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07 . 2

Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

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Properties causing ozone depletion

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07 . 3

Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

1

2

3

0 7 . 4 State what is meant by an **ozone hole**.

[1 mark]

0 7 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

0 7 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

15

0 8 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

- A A waste is assumed to cause pollution until research shows it is unlikely to do so ☐
- B The polluter must pay for any damage caused ☐
- C Emissions should be as low as reasonably achievable ☐
- D Only pollutants known to be toxic need to be controlled ☐

0 8 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

08 . 3

Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

1

2

5

09 . 1

Explain why it is difficult to be certain that a particular flood or heatwave is caused by anthropogenic climate change.

[3 marks]

09 . 2

Explain why historic temperature records collected in towns may overestimate global warming.

[2 marks]

09 . 3

Describe how each of the following sources of proxy data is used to estimate past climates and give one limitation.

[5 marks]

Tree rings (dendrochronology)

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Ice cores

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10

1 0

Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O ₂)	2
Triatomic oxygen / ozone (O ₃)	3

1 0

1

Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0 . 2

Give **two** other ways in which the atmosphere supports life.

[2 marks]

1

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2

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1 0 . 3

Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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1 0 . 4

The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

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Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2

Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

END OF QUESTIONS